Multi-stage Transfer Learning for Joint Classification and Segmentation of Chest X-ray Images

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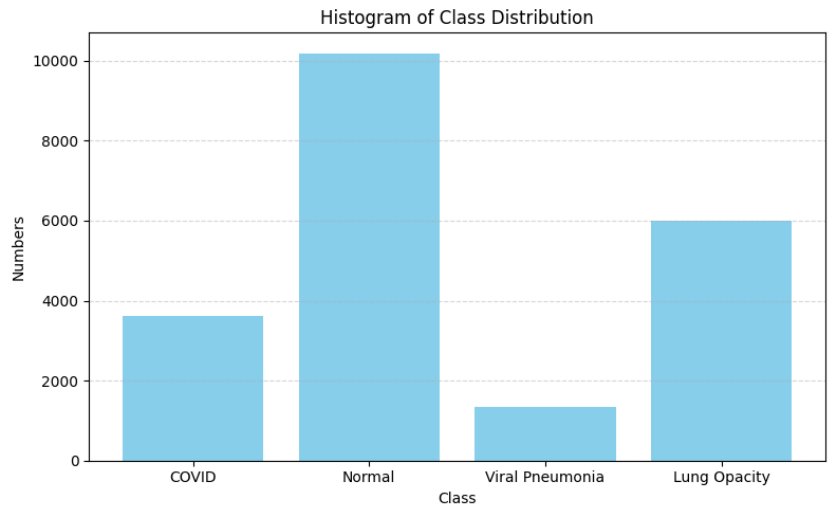
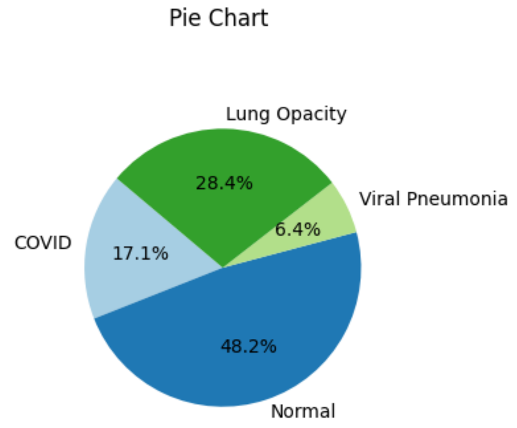
Motivation & Background

- Chest X-ray is a widely used tool for diagnosing lung diseases like COVID-19 and pneumonia.
- Manual diagnosis is time-consuming, error-prone, and inconsistent across radiologists.
- Single-task classification models struggle with complex backgrounds and ignore spatial structure like lung masks.
- Introducing lung segmentation as auxiliary information can guide the model to focus on key regions to improve classification accuracy

Dataset Overview + Class Distribution

- COVID-19 Radiography Dataset (Kaggle)
- The dataset contains **4 categories** of chest X-ray images:
- COVID-19, Normal, Viral Pneumonia, and Lung Opacity
- Each sample includes:
- A **chest X-ray image**
- A corresponding **lung mask** (grayscale)
- The **lung mask** highlights the anatomical region of the lungs
- It contains **no lesion annotations**
- Used as **structural prior** to help the model focus on relevant areas
- Class distribution is imbalanced (see charts →)



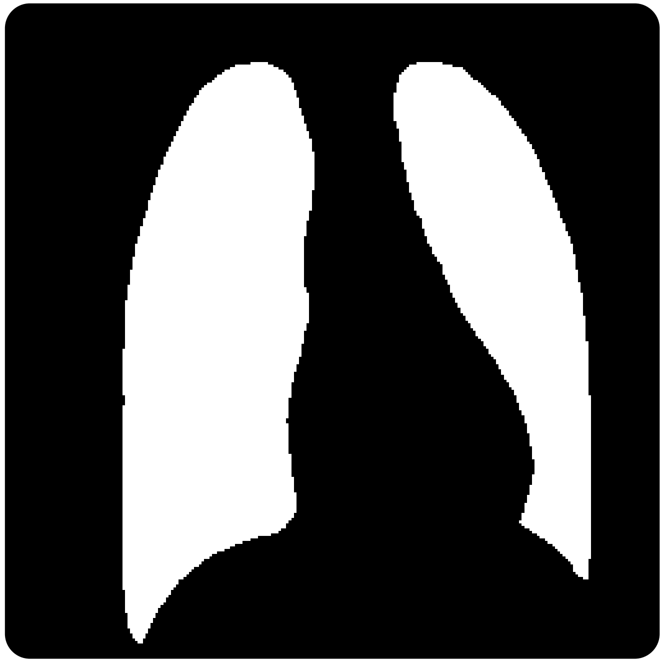


Handling Class Imbalance & Data Preparation

- **Stratified Data Splitting:**
- Ensures each class is proportionally represented in train, val, and test sets
- 70% training, 15% validation, 15% test
- Test set is **kept unseen** during training and only used for final evaluation
- **Data Augmentation (Training Set Only):**
- Random horizontal flip
- Random rotation (±10°)
- Resize to 224×224
- These techniques help improve generalization and robustness
- **Weighted Cross-Entropy Loss (Model Training):**
- Class weights are calculated based on training set distribution
- Helps reduce the bias toward over-represented classes

Original COVID-19 X-ray

Corresponding lung mask



Sample Images & Challenges in Chest X-ray Classification

Chest X-ray images often contain **complex background and noisy features**, such as bones, text, and medical artifacts.

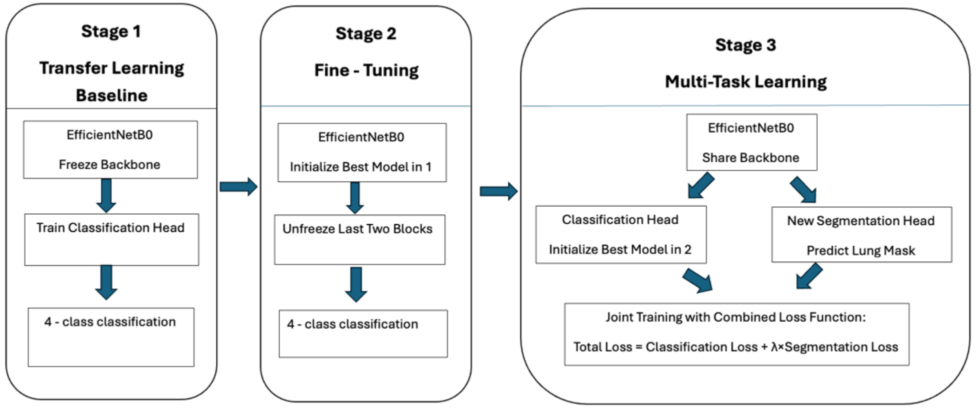
These elements can mislead classification models, especially in subtle cases.

Traditional models typically **lack spatial awareness** and treat the entire image equally.

Lung masks provide anatomical priors, helping the model focus on disease-relevant regions.

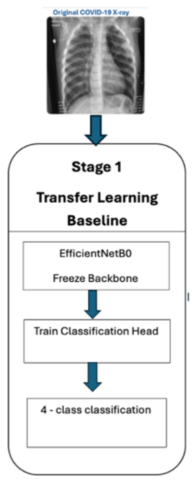
Model Design

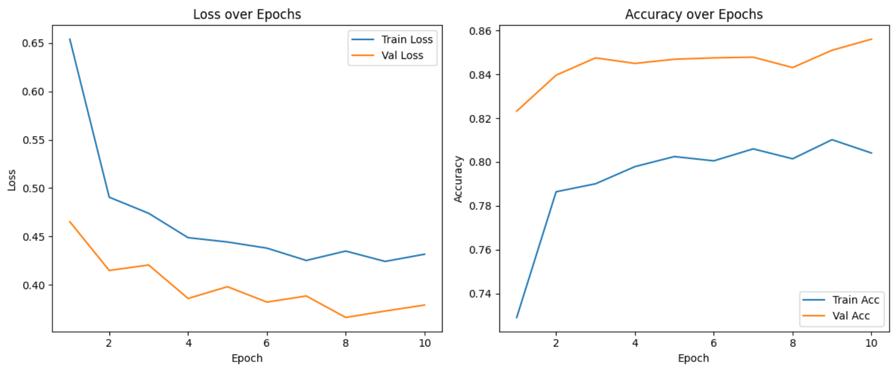
- **Three-Stage Training Pipeline**
- **Stage 1**: Build a baseline classifier using transfer learning
- → Freeze backbone, train classification head
- → Save best model
- **Stage 2**: Fine-tune best model from Stage 1
- → Unfreeze upper layers for domain adaptation
- → Save best model
- **Stage 3**: Construct multi-task model
- → Classification head from Stage 2
- → Add segmentation head to predict lung masks
- → Joint training improves accuracy and interpretability



Stage 1: Transfer Learning Baseline

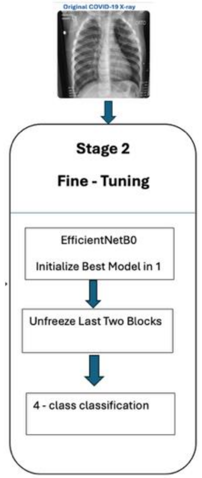
- **Backbone:** EfficientNetB0 pretrained on ImageNet
- **Strategy:** Freeze backbone, train only classification head
- **Task:** 4-class softmax classification
- **Training:** 10 epochs, Adam optimizer (LR = 0.001, batch size = 32)
- **Best model saved** for Stage 2 fine-tuning
- **Goal:** Establish a fast, stable performance benchmark
- Training and validation curves show **smooth convergence**

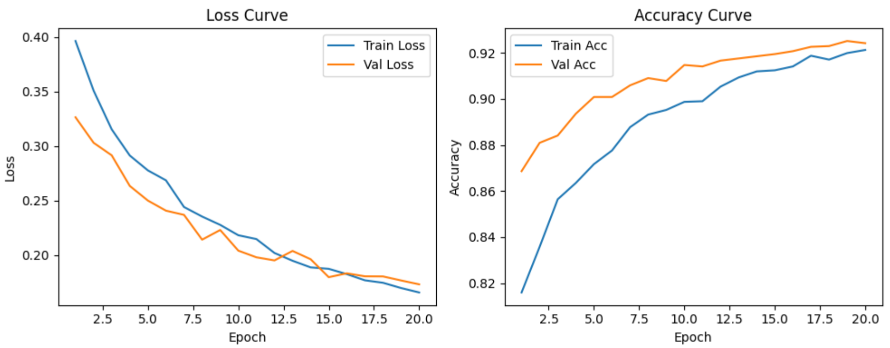




Stage 2: Fine-Tuning

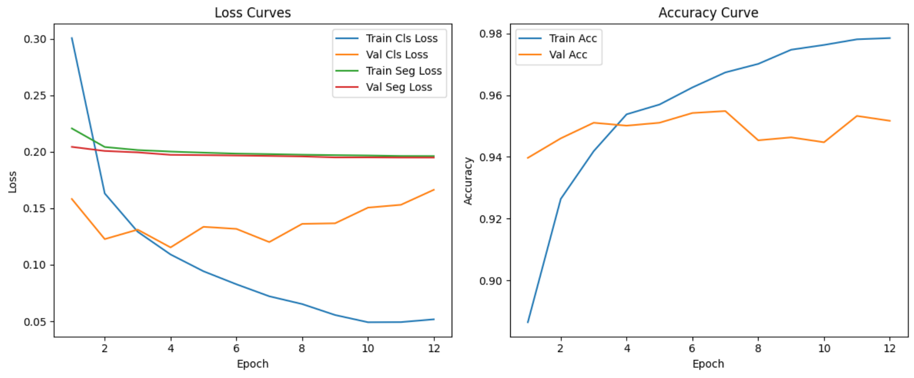
- Initialize from **Stage 1 best model**
- **Unfreeze last two blocks** (layers 6 and 7) of EfficientNetB0
- Allow partial adaptation to chest X-ray domain
- Use **ReduceLROnPlateau** scheduler to adjust learning rate
- Apply **Early Stopping** based on validation loss
- **Best model saved** for Stage 3 multi-task learning
- Trained for up to 20 epochs with batch size = 32

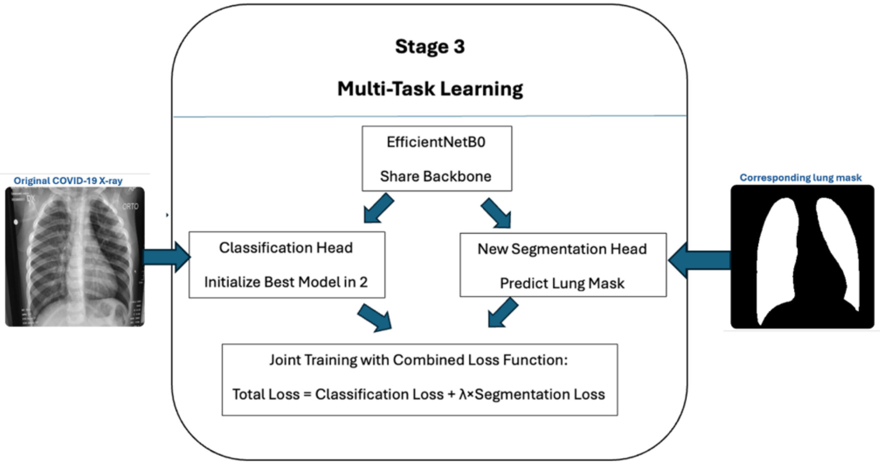


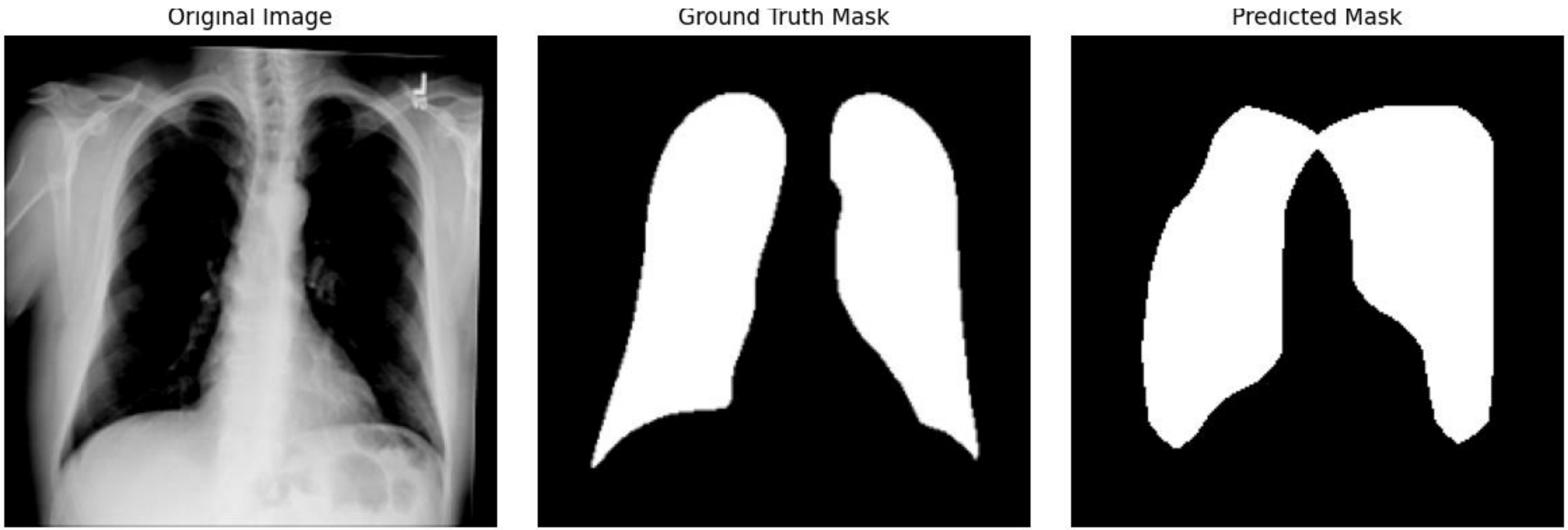


Stage 3: Multi-task Learning

- **Classification branch** initialized from Stage 2 best model
- Add a new **segmentation head** to predict lung masks
- Shared backbone (EfficientNetB0) used for both tasks
- **Segmentation loss:** Binary Cross-Entropy
- Joint training with **combined loss function**:
- Total Loss = Classification Loss + λ × Segmentation Loss
- Lung mask guides model to focus on relevant spatial regions
- Improves classification performance and interpretability
- Training curves show **stable optimization** of both tasks





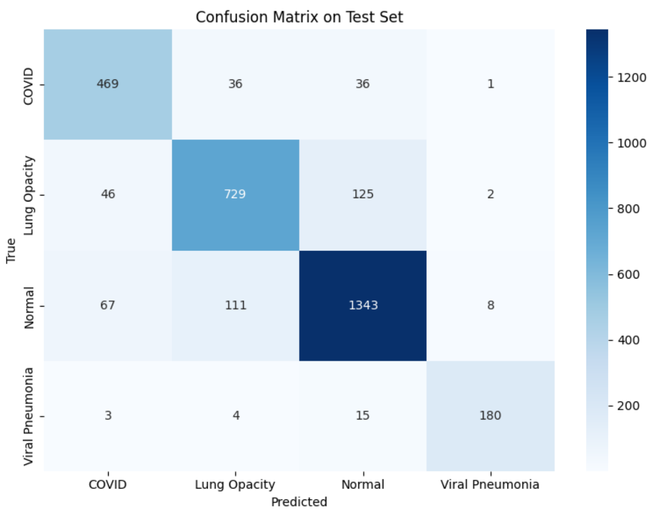


The predicted lung mask aligns well with the ground truth, indicating effective spatial learning from the segmentation task.

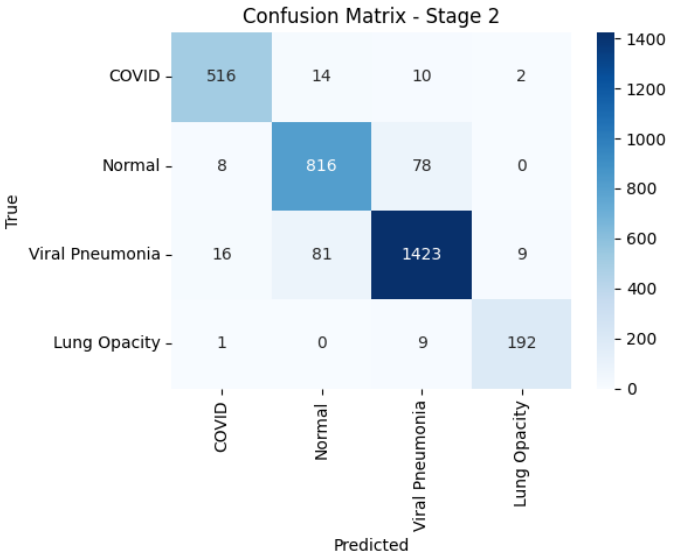
Stage 3 – Segmentation Output Visualization

- Model successfully learned anatomical lung structures
- Segmentation improves spatial attention and interpretability
- Supports classification by suppressing irrelevant background features

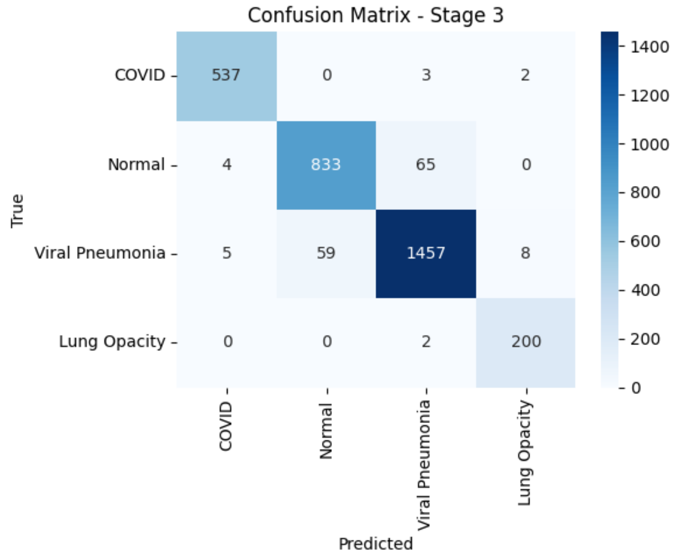
Confusion Matrix – Stage 1



Confusion Matrix – Stage 2



Confusion Matrix – Stage 3

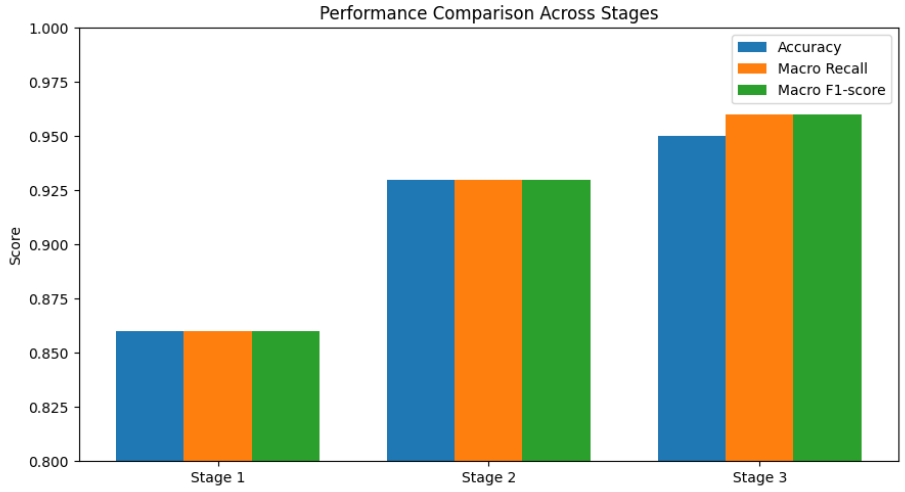


Test Set Results Comparison

- Stage 1 shows frequent misclassification between **Normal** and **Lung Opacity**
- Stage 2 improves recognition of **minority classes**, especially Long Opacity
- Stage 3 achieves **best balance** across all four categories
- Fewer false positives and false negatives across the board

Overall Performance Comparison

- All metrics show **consistent improvement** across stages
- Stage 2 fine-tuning significantly boosts recall and F1-score
- Stage 3 multi-task model achieves **highest performance and best balance**
- Classification becomes more reliable across all classes



Metric	Stage 1	Stage 2	Stage 3
Accuracy (%)	86.4	93.1	95.7
Macro Recall (%)	86.2	92.9	96.0
Macro F1-score (%)	85.7	93.2	96.1

Conclusion

- Developed a three-stage deep learning pipeline for chest X-ray classification
- Combined classification and lung segmentation in a **multi-task model**
- Achieved **95.7% test accuracy** and **96.1% macro F1-score**
- Multi-task learning improves both performance and interpretability
- Demonstrated the value of segmentation as a structural prior in medical imaging

Thank You